Assignment: Image Segmentation

Task-1:

For each scenario, select **one or more architectures that could be investigated**, and justify your selection using architectural characteristics rather than simply naming a model.

- **Scenario A**
  You need to segment very small anatomical structures where accurate boundaries are important.

- **Scenario B**
  The target objects appear at widely different scales.

- **Scenario C**
  The image contains substantial irrelevant background information.

- **Scenario D**
  The segmentation task requires strong global/contextual information.

- **Scenario E**
  You have limited computational resources and need a relatively simple segmentation architecture.

Important: Your answer must explain the architectural reasoning. Do not simply write the model name.

Task-2:

A. Train the following segmentation models on the **same image-segmentation dataset** such as Oxford-IIIT Pet dataset.

   - U-Net
   - U-Net++
   - FPN-based Segmenter
   - Attention U-Net
   - BASNet
   - DeepLabV3+
   - SegFormer

   For each model, report the following:

   - Dataset description

- Number of images
- Image resolution
- Number of segmentation classes
- Training/validation/test split
- Preprocessing
- Data augmentation
- Loss function
- Optimizer
- Learning rate
- Number of epochs
- Batch size
- Hardware used

B. Evaluate using: IoU, Dice, Pixel Accuracy, Precision, Recall, F1-score, Mean IoU.

C. Present the results in a **comparative table** and discuss the differences in performance among the models.

Task-3:

Design your own customized image-segmentation model by modifying, combining, or extending components of existing segmentation architectures.

Your customized model should include at least **one meaningful architectural modification**, such as:

- A modified encoder or decoder
- A new or modified skip-connection strategy
- An attention mechanism
- A multi-scale feature-fusion mechanism
- A different upsampling method
- A boundary-refinement module
- A modified loss function
- A combination of components from different segmentation architectures

You must:

A. Describe the motivation behind your proposed architecture.

B. Draw and explain the complete architecture of your customized model.

C. Clearly identify which components are adopted from existing models and which components are your own modifications.

D. Train your customized model using the **same dataset and experimental settings** used for the other models, as far as practicable.

E. Evaluate it using the same evaluation metrics.

F. Compare its performance with the other popular segmentation models.

G. Discuss the advantages, limitations, and possible reasons for its performance.

Task-4:

A. Capture **several images using your mobile phone** that are relevant to the segmentation task.

B. Manually annotate the images to create their **ground-truth segmentation masks**.

C. Apply the same preprocessing used for the test dataset to these images.

D. Use each trained model to predict segmentation masks for the self-captured images.

E. Compare the predicted masks with your manually annotated ground-truth masks.

F. Calculate the same evaluation metrics used for the test dataset.

G. Display, for each selected image:
Input Image → Ground-Truth Mask → Predicted Mask

H. Discuss the performance of the models on the self-captured images and identify possible reasons for any performance differences compared with the original test dataset.